

Triangle

Authored by: Arpit Raj, LNMIIT Jaipur

Date: September 19, 2026

Target Companies: DE Shaw, Agoda, Goldman Sachs, Salesforce

Problem Visualization

```
      2
     3 4
    6 5 7
   4 1 8 3
```

Rule: can move to either i or $i+1$ of next

here

[2 + 3 + 5 + 1 = 11]

or

[2 + 4 + 5 + 1 = 12]

similarly there are many traversal options

DP State & Transitions

Basically:

```
dp[0][0] = arr[0][0]
```

then

$dp[r][c]$

$dp[r+1][c]$

$dp[r+1][c+1]$

```
nr = r + 1
nc1 = c
nc2 = c + 1
```

```
dp[nr][nc1] = min(dp[nr][nc1], dp[r][c] + arr[nr][nc1])
// similar for dp[nr][nc2]
```

and then take min from bottom row

C++ Implementation (lc_120)

```
class Solution { public: int minimumTotal(vector<vector<int>>& triangle)
{ int n=triangle.size(); vector<vector<int>> dp(n); for(int i=0;i<n;i++){
dp[i]=vector<int>(triangle[i].size(),INT_MAX); } dp[0][0]=triangle[0][0];
for(int r=0;r<n-1;r++){ for(int c=0;c<=r;c++){ int nr=r+1; int nc1=c; int
nc2=c+1; dp[nr][nc1]=min(dp[nr][nc1],dp[r][c]+triangle[nr][nc1]); dp[nr]
[nc2]=min(dp[nr][nc2],dp[r][c]+triangle[nr][nc2]); } } int ans=INT_MAX;
for(int col=0;col<n;col++){ ans=min(ans,dp[n-1][col]); } return ans; } };
```